For example : Gierer- Meinhardt model (1982) :

$$\begin{cases} \dfrac{\partial a}{\partial t} = \dfrac{c_1 a^2}{h} - \mu a + D_a \nabla^2 a & : \text{activator} \\[2ex] \dfrac{\partial h}{\partial t} = c_2 a^2 - \nu h + D_h \nabla^2 h & : \text{inhibitor} \end{cases}$$

A homogeneous steady state $(\bar{a}, \bar{h})$ satisfies:

$$\begin{cases} \dfrac{c_1 \bar{a}^2}{\bar{h}} - \mu \bar{a} = 0 \\[2ex] c_2 \bar{a}^2 - \nu \bar{h} = 0 \end{cases} \Rightarrow \begin{cases} \bar{a} = \dfrac{c_1 \nu}{c_2 \mu} \\[2ex] \bar{h} = \dfrac{c_1^2 \nu}{c_2 \mu^2} \end{cases}$$

The Jacobian for the system is :

$$J = \begin{pmatrix} \partial R_1/\partial a & \partial R_1/\partial h \\ \partial R_2/\partial a & \partial R_2/\partial h \end{pmatrix}\Bigg|_{\bar{a},\bar{h}} = \begin{pmatrix} \dfrac{2c_1 a}{h} - \mu & \dfrac{-c_1 a^2}{h^2} \\[2ex] 2c_2 a & -\nu \end{pmatrix}\Bigg|_{\bar{a},\bar{h}} = \begin{pmatrix} \underset{a_{11}}{\mu} & -\mu^2/c_1 \\[2ex] \dfrac{2c_1\nu}{\mu} & \underset{a_{22}}{-\nu} \end{pmatrix}$$

Condition 3 becomes : $\mu D_h - \nu D_a > 2(\mu\nu)^{1/2}(D_a D_h)^{1/2}$

OR

$$\sqrt{\frac{D_h \mu}{D_a \nu}} > 1 + \sqrt{2}.$$

Admits spots only :(

A common way to obtain stripe-like patterns in this model is to consider a self-activation term : $\dfrac{a^2}{h} \rightarrow \dfrac{a^2}{h(1+Ka^2)}$ $K>0$ is a saturation constant

$$\begin{cases} \dfrac{\partial a}{\partial t} = \dfrac{c_1 a^2}{h(1+Ka^2)} - \mu a + D_a \nabla^2 a \\[2ex] \dfrac{\partial h}{\partial t} = c_2 a^2 - \nu h + D_h \nabla^2 h \end{cases}$$